

7 **Ans: A**

Initially,

$F = kx$ for each wire

$\frac{1}{3} W = kx$ (Weight of lamp W is shared equally by the 3 wires)

$$W = 3kx$$

Finally,

$\frac{1}{2} W = kx'$ (W is now shared equally by the 2 wires)

$$\frac{1}{2} (3 kx) = kx'$$

$$x' = \frac{3}{2} x$$

difference between $h = x' - x = \frac{3}{2} x - x = 0.5 x = 0.5 (0.4 \text{ cm}) = 0.20 \text{ cm}$

8 **Ans: A**